OSI Model Layer 1 – Physical Layer

Layer 1 is responsible for physically moving bits from one device to another. does not care whether the data represents an email, image, video, or webpage. It is concerned with how the binary code (1s and 0s) physically travel.

Depending on the technology, the signal may be represented using:

- Electrical signals
- Light pulses
- Radio waves

First responsibility is bit transmission, it converts bits into signals that can travel through a medium.

Second responsibility is how will the devices communicates and which physical medium will be used.

Examples of physical mediums are:

- Copper Cable - Cat5e, Cat6, Cat6a, and Cat7
- Fiber Optic Cable
- Wi-Fi
- Bluetooth
- Cellular communication

**Note: some of the examples of physical mediums are intangible? Yes, in the context of OSI Model the physical mediums deals with how the signals travel not how humans physically interact with.*

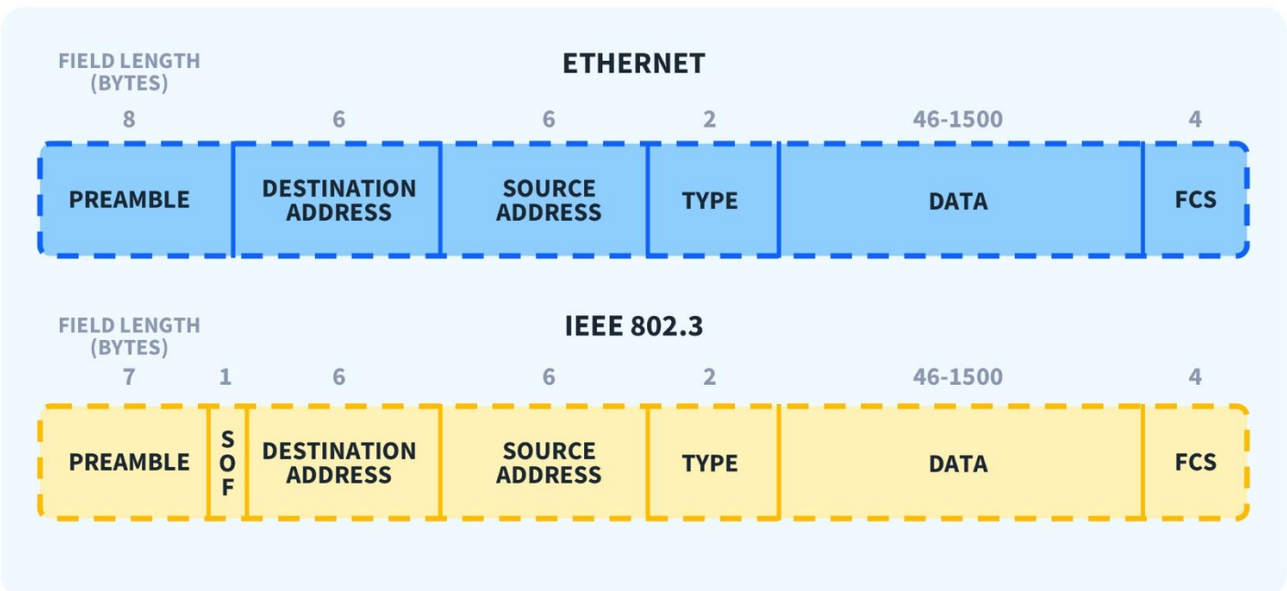
Feature	Copper	Fiber	Wireless
Signal	Electrical	Light	Radio
Physical cable	Yes	Yes	No
EMI resistance	Lower	Very high	Can experience interference
Typical use	LANs	Backbone/long distance	Wi-Fi/mobile
Installation	Relatively easy	More specialized	No cable to endpoint
Distance	Generally shorter	Generally longer	Depends on environment

OSI Model Layer 2 – Data Link Layer

The Layer 2 is responsible for providing communication between devices that are directly connected or operating on the same local network. In this layer the received bits from the signals in the Layer 1 groups those bits into something called a **frame**.

A **frame** is a structured unit of data used at Layer 2.

Simple analogy: It's like an envelope containing information about where the data came from and where it should go**.**



The components of a Frame are:

- Preamble – used to help the receiving device synchronize its clock with the incoming Ethernet signal.
- Start Frame Delimiter – this signals that the ethernet frame starts now.
- Destination MAC address – The physical address of the end device to receive the data.
- Source MAC address – The physical address of the end device that sent the data.
- EtherType – It tells the receiver how much data is contained in the frame's payload.
- Data – the actual content of the frame.
- Frame check information – this checks if the received bits were in a sequential manner.

The Data Link Layer (Layer 2) is divided into two sublayers: **LLC – Logical Link Control** and the **MAC – Media Access Control** they have different responsibilities. **LLC sublayer** is responsible mainly for providing a connection between the Data Link Layer and the Network Layer. **MAC sublayer** is more directly involved in the actual local network delivery of frames from MAC addressing, framing, access to the transmission medium, and frame integrity mechanisms.

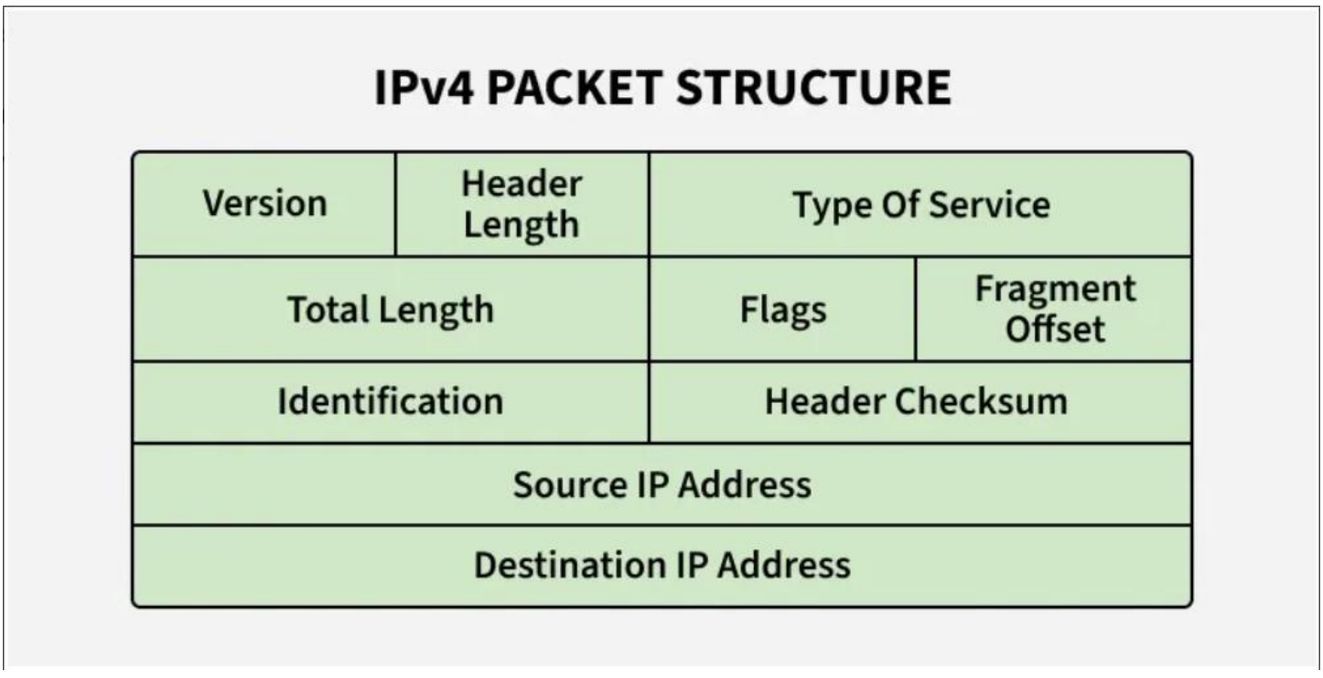
**Note: The exact division of responsibilities can vary by link-layer technology. For example, Ethernet and Wi-Fi implement MAC functions differently.*

OSI Model Layer 3 – Network Layer

Its primary job is to provide **logical addressing and routing** so that data can travel from one network to another. ***Layer 2 vs Layer 3:*** in Layer 2 it focuses on **local delivery** while in Layer 3 it focuses on **communication between networks**.

In the Network Layer the data unit is called a **packet.** The IP packet contains important information such as:

- Source IP address
- Destination IP address
- TTL
- Protocol
- Other control information



How Octets Work

A full IPv4 address contains **4 octets** separated by dots, totaling **32 bits**:

Dotted Decimal Format: 192.168.1.1
Binary Format: 11000000.10101000.00000001 00000001
Structure: [1st Octet].[2nd Octet].[3rd Octet].[4th Octet]

Each individual octet can have a decimal value ranging from **0 to 255** (which is the decimal equivalent of 00000000 to 11111111 in binary).

Metric	Value	Description
Subnet Mask	255.255.255.0	The 32-bit mask separating network and host.
CIDR Notation	/24	Shortened way to say "24 network bits".
Total IP Addresses	256	Total combinations in the 4th octet (2^8).
Usable IP Addresses	254	Total IPs you can actually assign to devices (2^8 - 2).

In every subnet, you must subtract 2 addresses from the total count because they have reserved, unassignable jobs:

The Network ID (.0): The very first address. It names and identifies the entire network.

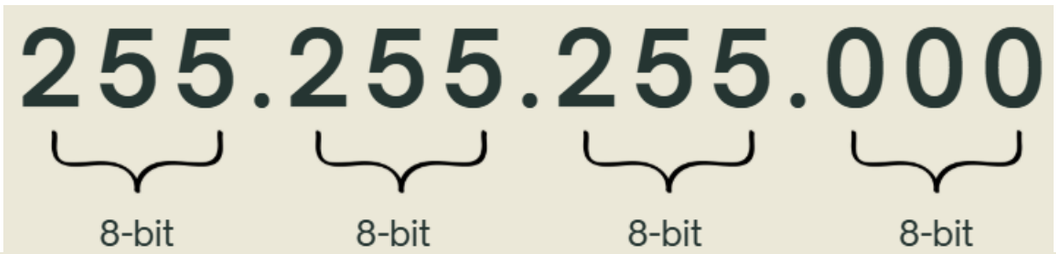
The Broadcast Address (.255): The very last address. It is used to send data to every device on the network simultaneously.

Example:

Network ID: 192.168.1.0
First Usable IP: 192.168.1.1 (usually assigned to your router/gateway)
Last Usable IP: 192.168.1.254 (assigned to computers, phones, printers)
Broadcast Address: 192.168.1.255

CIDR Notation

Classless Inter-Domain Routing (CIDR) uses a slash "/" followed by a number to show how many bits make up the network portion.



To calculate the usable host addresses use the formula below.

x = Total bit of Network Portion y = Total bit of Host Portion

$y = (x - 32)$

Total usable Host Portion = 2^y

For example, a /24 CIDR notation:

$y = (24 - 32)$: $y = 8$
Total usable Host Portion = 2^8 : **Total usable Host Portion = 256**

/24 means the first 24 bits are for the network, leaving 8 bits for devices.
/24 gives 256 total addresses per subnet.

Core Functions of Layer 3 IP Packets

Logical Addressing: Uses unique IP addresses (IPv4 or IPv6) to identify the source and destination devices globally across the internet.

Routing: Directs packets along the best path across different networks

Encapsulation: Wraps Layer 4 data (TCP/UDP segments) with an IP header containing source and destination IPs, a protocol identifier, and a Time-to-Live (TTL)

Fragmentation: Breaks down large packets into smaller pieces if the destination link has a smaller Maximum Transmission Unit (MTU).

DHCP stands for **Dynamic Host Configuration Protocol**.

It is a network protocol that automatically gives devices the network information they need to communicate on a network. Without DHCP, you would have to manually configure things like IP address, Subnet mask, Default gateway, and DNS server. DHCP is mainly associated with Layer 7 – Application Layer because it is an application-layer protocol.

Simple definition: DHCP automatically assigns IP configuration to devices on a network.

Why do we need an **DHCP?**

When handling large networks this would be extremely difficult and could cause duplicate IP addresses, incorrect subnet masks, wrong gateways, incorrect DNS settings Configuration.

There are 2 important roles in DHCP it is **DHCP Server** and **DHCP Client**. The **DHCP Server** provides the network configuration such as automatically leasing IP addresses, configuring network subnets and default gateways, and avoiding IP address conflicts. While the **DHCP Client** are network End Devices that requests and receives its IP address and configuration settings automatically from a DHCP Server.

DHCP Server = Leases out IP addresses to end devices that needs one.
DHCP Client = Requests an IP address to the DHCP Server.

The DHCP Process

Discover → Offer → Request → Acknowledge

1. **DHCP Discover** – the client does not have an IP address yet and sends out a broadcast to find a DHCP Server.
2. **DHCP Offer** – a DHCP server receives the request and offers an IP address.
3. **DHCP Request** – the client requests the offered configuration.
4. **DHCP Acknowledgment** – the server confirms the assignment then the client can now configure its network interface.

DHCP Pool – a DHCP Pool is a range of addresses that the DHCP server can assign.

What is a **Network Port?**

A **network port** is a logical number used by TCP or UDP to identify a particular service or application on a device. Think of an IP address as identifying the device, while a port identifies the service/application on that device.

Examples of Common Ports:

Port	Protocol	Service
20	TCP	FTP Data
21	TCP	FTP Control
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	TCP/UDP	DNS
67	UDP	DHCP Server
68	UDP	DHCP Client
80	TCP	HTTP
110	TCP	POP3
143	TCP	IMAP
161	UDP	SNMP
389	TCP/UDP	LDAP
443	TCP	HTTPS
445	TCP	SMB
3389	TCP	RDP